

## 1. General Description

The EM74CBTLV3125 provides a 4-bit high-speed bus switch with separate output enable inputs ( $\overline{1OE}$  to  $\overline{4OE}$ ). The low on-state resistance of the switch allows connections to be made with minimal propagation delay. The switch is disabled (high-impedance OFF-state) when the output enable ( $\overline{nOE}$ ) input is HIGH.

To ensure the high-impedance OFF-state during power-up or power-down,  $\overline{nOE}$  should be tied to the  $V_{CC}$  through a pull-up resistor. The minimum value of the resistor is determined by the current-sinking capability of the driver.

Schmitt trigger action at control input makes the circuit tolerant to slower input rise and fall times across the entire  $V_{CC}$  range from 2.3 V to 3.6 V.

This device is fully specified for partial power-down applications using  $I_{OFF}$ . The  $I_{OFF}$  circuitry disables the output, preventing the damaging backflow current through the device when it is powered down.

## 2. Features and Benefits

- Supply voltage range from 2.3 V to 3.6 V
- 2.4  $\Omega$  switch connection between two ports
- CMOS low power dissipation
- $I_{OFF}$  circuitry provides partial Power-down mode operation
- Latch-up performance exceeds 250 mA
- Rail to rail switching on data I/O ports
- Complies with JEDEC standard:
  - JESD8-5 (2.3 V to 2.7 V)
  - JESD8B/JESD36 (2.7 V to 3.6 V)
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 6000 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

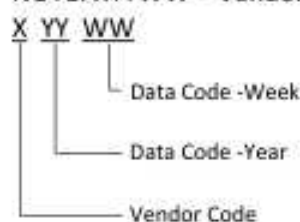
## 3. Ordering Information

Table 1. Ordering information

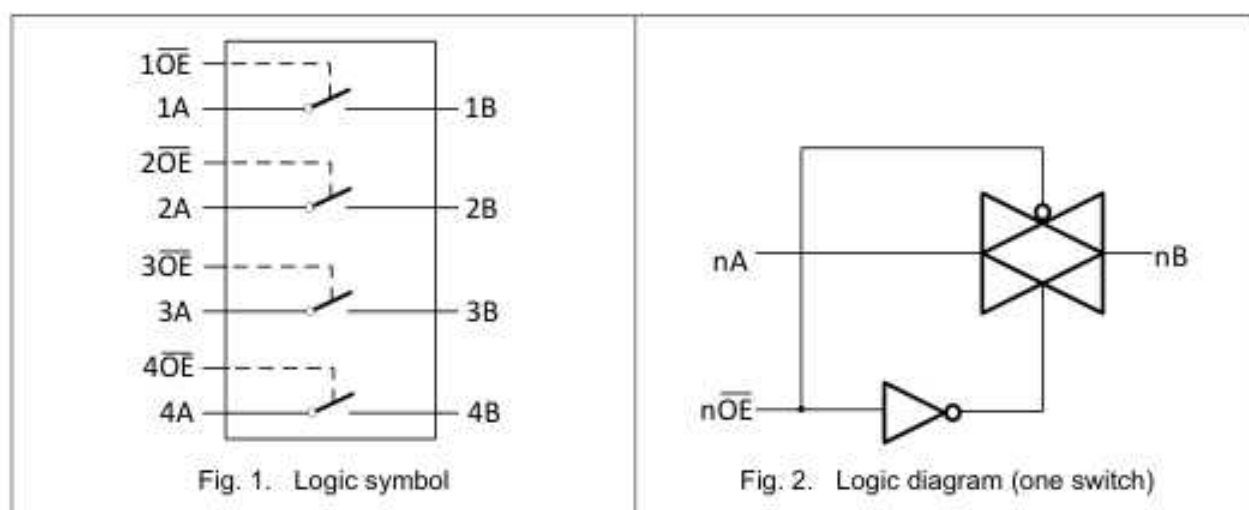
| Type number     | Topside marking    | Package   |                                                                           |          |
|-----------------|--------------------|-----------|---------------------------------------------------------------------------|----------|
|                 |                    | Name      | Description                                                               | Quantity |
| EM74CBTLV3125D  | CBTLV3125<br>XYYWW | SOP-14L   | plastic small outline package; 14 leads;<br>body width 3.9 mm             | 3000     |
| EM74CBTLV3125PW | CBTLV3125<br>XYYWW | TSSOP-14L | plastic thin shrink small outline package; 14<br>leads; body width 4.4 mm | 3000     |

### MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Data Code.

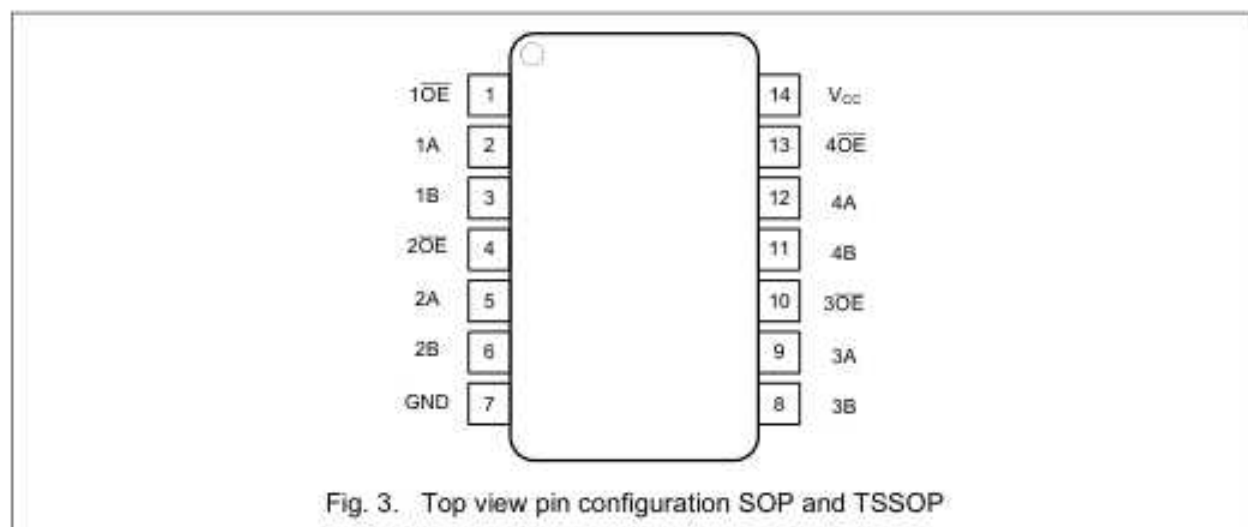


## 4. Function Diagram



## 5. Pinning Information

### 5.1. Pinning



### 5.2. Pin description

Table 2. Pin description

| Symbol                                                                    | Pin          | Description             |
|---------------------------------------------------------------------------|--------------|-------------------------|
| $\overline{1OE}$ , $\overline{2OE}$ , $\overline{3OE}$ , $\overline{4OE}$ | 1, 4, 10, 13 | Output enable input     |
| 1A, 2A, 3A, 4A                                                            | 2, 5, 9, 12  | A input/output          |
| 1B, 2B, 3B, 4B                                                            | 3, 6, 8, 11  | B output/input          |
| GND                                                                       | 7            | Ground (0 V)            |
| V <sub>CC</sub>                                                           | 14           | Positive supply voltage |

## 6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

| Output enable input $\overline{OE}$ | Function switch |
|-------------------------------------|-----------------|
| L                                   | ON-state        |
| H                                   | OFF-state       |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                      | Min  | Max          | Unit |
|-----------|-------------------------|---------------------------------|------|--------------|------|
| $V_{CC}$  | supply voltage          |                                 | -0.5 | 4.6          | V    |
| $I_{IK}$  | input clamping current  | $V_I < -0.5$ V                  | -50  |              | mA   |
| $V_I$     | input voltage           | control inputs [1]              | -0.5 | 4.6          | V    |
| $V_{SW}$  | switch voltage          | enable and disable mode [2]     | -0.5 | $V_{CC}+0.5$ | V    |
| $I_{SK}$  | switch clamping current | $V_I < -0.5$ V                  | -50  |              | mA   |
| $I_{SW}$  | switch current          | $V_{SW} = 0$ V to $V_{CC}$      |      | $\pm 128$    | mA   |
| $I_{CC}$  | supply current          |                                 |      | 100          | mA   |
| $I_{GND}$ | ground current          |                                 | -100 |              | mA   |
| $P_{tot}$ | total power dissipation | $T_{amb} = -40$ °C to $+125$ °C |      | 500          | mW   |
| $T_{stg}$ | storage temperature     |                                 | -65  | 150          | °C   |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

[2] The switch voltage ratings may be exceeded if switch clamping current ratings are observed

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions                         | Min | Max      | Unit |
|---------------------|-------------------------------------|------------------------------------|-----|----------|------|
| $V_{CC}$            | supply voltage                      |                                    | 2.3 | 3.6      | V    |
| $V_I$               | input voltage                       | control inputs                     | 0   | 3.6      | V    |
| $V_{SW}$            | switch voltage                      | enable and disable mode            | 0   | $V_{CC}$ | V    |
| $T_{amb}$           | ambient temperature                 |                                    | -40 | 125      | °C   |
| $\Delta t/\Delta V$ | input transition rise and fall rate | pin nOE; $V_{CC} = 2.3$ V to 3.6 V | 0   | 200      | ns/V |

## 9. Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

| Symbol              | Parameter                 | Conditions                                                                                                                                            | -40 °C to +85 °C |        |     | -40 °C to +125 °C |      | Unit |
|---------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------|-----|-------------------|------|------|
|                     |                           |                                                                                                                                                       | Min              | Typ[1] | Max | Min               | Max  |      |
| V <sub>IH</sub>     | HIGH-level input voltage  | V <sub>CC</sub> = 2.3 V to 2.7 V                                                                                                                      | 1.7              |        |     | 1.7               |      | V    |
|                     |                           | V <sub>CC</sub> = 3.0 V to 3.6 V                                                                                                                      | 2.0              |        |     | 2.0               |      | V    |
| V <sub>IL</sub>     | LOW-level input voltage   | V <sub>CC</sub> = 2.3 V to 2.7 V                                                                                                                      |                  |        | 0.7 |                   | 0.7  | V    |
|                     |                           | V <sub>CC</sub> = 3.0 V to 3.6 V                                                                                                                      |                  |        | 0.9 |                   | 0.9  | V    |
| I <sub>I</sub>      | input leakage current     | pin $\overline{\text{nOE}}$ ; V <sub>CC</sub> = 3.6 V ;<br>V <sub>I</sub> = GND to V <sub>CC</sub>                                                    |                  |        | ±1  |                   | ±20  | μA   |
| I <sub>S(OFF)</sub> | OFF-state leakage current | V <sub>CC</sub> = 3.6 V ; see Fig. 4                                                                                                                  |                  |        | ±1  |                   | ±20  | μA   |
| I <sub>S(ON)</sub>  | ON-state leakage current  | V <sub>CC</sub> = 3.6 V ; see Fig. 5                                                                                                                  |                  |        | ±1  |                   | ±20  | μA   |
| I <sub>OFF</sub>    | power-off leakage current | V <sub>CC</sub> = 0 V ;<br>V <sub>I</sub> or V <sub>O</sub> = 0 V to 3.6 V                                                                            |                  |        | ±10 |                   | ±50  | μA   |
| I <sub>CC</sub>     | supply current            | V <sub>CC</sub> = 3.6 V ;<br>V <sub>I</sub> = V <sub>CC</sub> or GND ; V <sub>SW</sub> = V <sub>CC</sub><br>or GND ; I <sub>O</sub> = 0 A ;           |                  |        | 10  |                   | 50   | μA   |
| ΔI <sub>CC</sub>    | additional supply current | pin $\overline{\text{nOE}}$ ; V <sub>CC</sub> = 3.6 V ; [2]<br>V <sub>I</sub> = V <sub>CC</sub> - 0.6 V ;<br>V <sub>SW</sub> = GND or V <sub>CC</sub> |                  |        | 300 |                   | 2000 | μA   |
| C <sub>I</sub>      | input capacitance         | pin $\overline{\text{nOE}}$ ; V <sub>CC</sub> = 3.3 V ;<br>V <sub>I</sub> = 0 V to 3.3 V                                                              |                  | 1.5    |     |                   |      | pF   |
| C <sub>S(OFF)</sub> | OFF-state capacitance     | V <sub>CC</sub> = 3.3 V ;<br>V <sub>I</sub> = 0 V to 3.3 V                                                                                            |                  | 5.2    |     |                   |      | pF   |
| C <sub>S(ON)</sub>  | ON-state capacitance      | V <sub>CC</sub> = 3.3 V ;<br>V <sub>I</sub> = 0 V to 3.3 V                                                                                            |                  | 12     |     |                   |      | pF   |

[1] All typical values are measured at T<sub>amb</sub> = 25 °C.

[2] One input at 3 V, other inputs at V<sub>CC</sub> or GND.

**Table 7. Resistance  $R_{ON}$** 

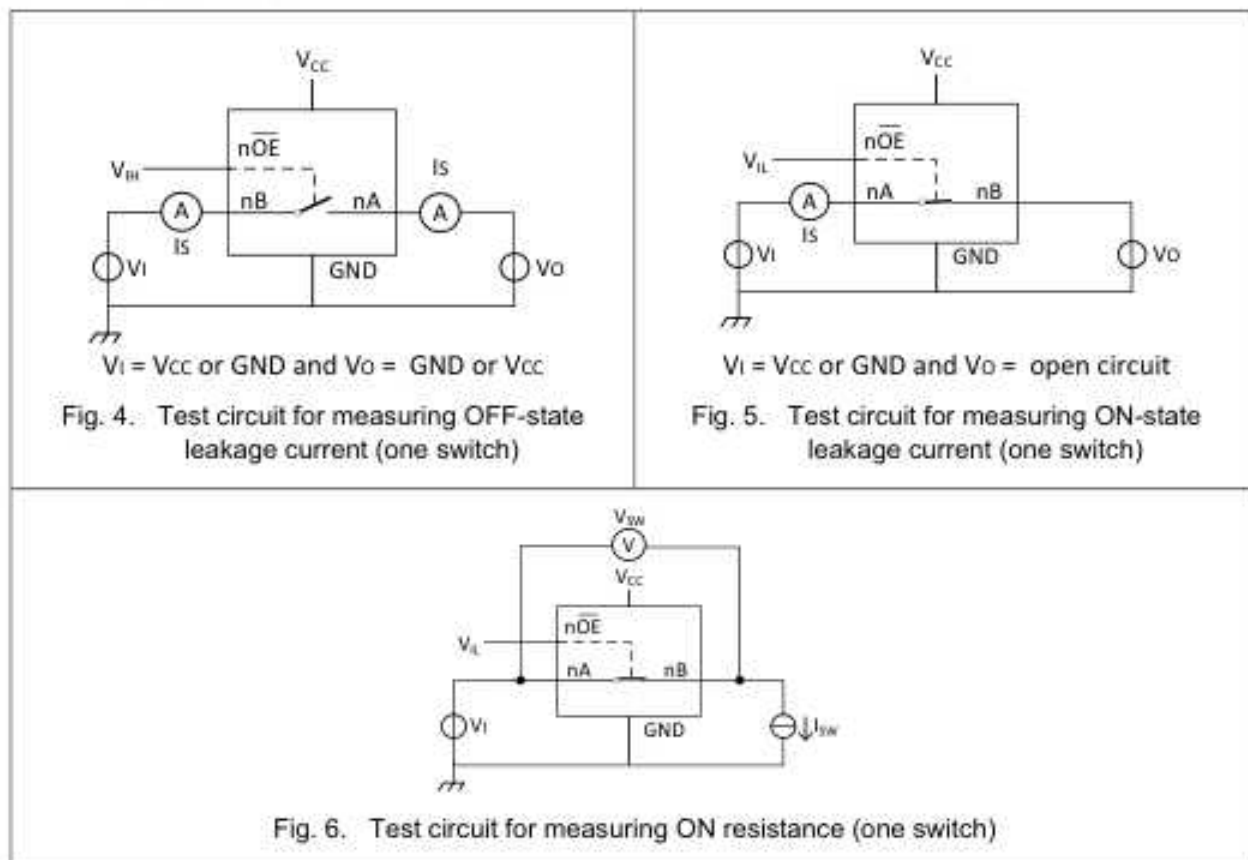
At recommended operating conditions; voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6.

| Symbol   | Parameter     | Conditions                                      | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit     |
|----------|---------------|-------------------------------------------------|------------------|--------|-----|-------------------|-----|----------|
|          |               |                                                 | Min              | Typ[1] | Max | Min               | Max |          |
| $R_{ON}$ | ON resistance | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V};$ [2] |                  |        |     |                   |     |          |
|          |               | $I_{SW} = 64 \text{ mA}; V_I = 0 \text{ V}$     |                  | 2.8    | 5   |                   | 6   | $\Omega$ |
|          |               | $I_{SW} = 24 \text{ mA}; V_I = 0 \text{ V}$     |                  | 2.8    | 5   |                   | 6   | $\Omega$ |
|          |               | $I_{SW} = 15 \text{ mA}; V_I = 1.7 \text{ V}$   |                  | 6.6    | 10  |                   | 12  | $\Omega$ |
|          |               | $V_{CC} = 3.0 \text{ V to } 3.6 \text{ V};$     |                  |        |     |                   |     |          |
|          |               | $I_{SW} = 64 \text{ mA}; V_I = 0 \text{ V}$     |                  | 2.5    | 4.5 |                   | 5.5 | $\Omega$ |
|          |               | $I_{SW} = 24 \text{ mA}; V_I = 0 \text{ V}$     |                  | 2.5    | 4.5 |                   | 5.5 | $\Omega$ |
|          |               | $I_{SW} = 15 \text{ mA}; V_I = 2.4 \text{ V}$   |                  | 4.4    | 8   |                   | 10  | $\Omega$ |

[1] All typical values are measured at nominal  $V_{CC}$  and  $T_{amb} = 25 \text{ }^\circ\text{C}$ .

[2] Measured by the voltage drop between the A and B terminals at the indicated current through the switch. ON-state resistance is determined by the lower of the voltages of the two (A or B) terminals.

## 9.1. Test circuit



## 10. Dynamic Characteristics

**Table 8. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 9.

| Symbol       | Parameter                | Conditions                                      | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|--------------|--------------------------|-------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|              |                          |                                                 | Min              | Typ[1] | Max | Min               | Max |      |
| $t_{pd}$     | propagation delay        | nA to nB or nB to nA; see Fig. 7 [2][3]         |                  |        |     |                   |     |      |
|              |                          | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$      |                  |        | 0.3 |                   | 0.4 | ns   |
|              |                          | $V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$      |                  |        | 0.3 |                   | 0.4 | ns   |
| $t_{en}$     | enable time              | $\overline{nOE}$ to nA or nB; see Fig. 8 [4]    |                  |        |     |                   |     |      |
|              |                          | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$      | 1.0              | 3.1    | 4.5 | 1.0               | 5.5 | ns   |
|              |                          | $V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$      | 1.0              | 2.8    | 4.2 | 1.0               | 5.3 | ns   |
| $t_{dis}$    | disable time             | $\overline{nOE}$ to nA or nB; see Fig. 8 [5]    |                  |        |     |                   |     |      |
|              |                          | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$      | 1.0              | 3      | 5.1 | 1.0               | 6   | ns   |
|              |                          | $V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$      | 1.0              | 2.9    | 5   | 1.0               | 6   | ns   |
| $f_{ -3dB }$ | -3 dB frequency response | $V_{CC} = 3.3 \text{ V}; R_L = 50 \Omega$ ; [6] |                  | 240    |     |                   |     | MHz  |

[1] All typical values are measured at  $T_{amb} = 25 \text{ °C}$  and at nominal  $V_{CC}$ .

[2] The propagation delay is the calculated RC time constant of the typical on-state resistance of the switch and the load capacitance, when driven by an ideal voltage source (zero output impedance).

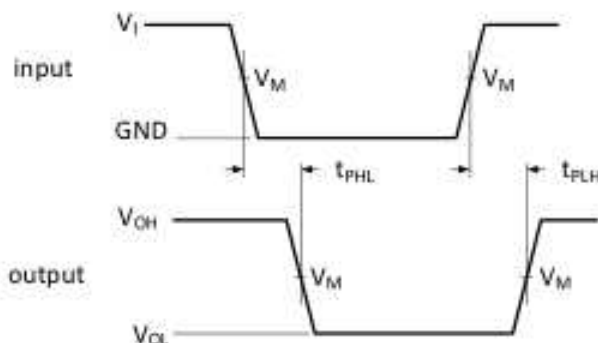
[3]  $t_{pd}$  is the same as  $t_{PLH}$  and  $t_{PHL}$ .

[4]  $t_{en}$  is the same as  $t_{ZOH}$  and  $t_{OL}$ .

[5]  $t_{dis}$  is the same as  $t_{FQ}$  and  $t_{QZ}$ .

[6]  $f_i$  is biased at  $0.5 \times V_{CC}$ .

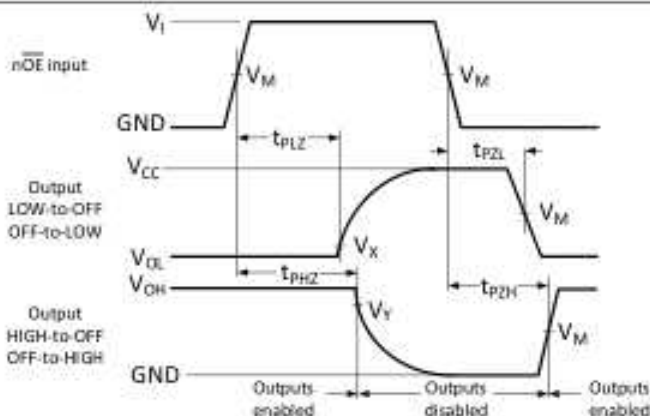
## 10.1. Waveforms and test circuit



Measurement points are given in Table 9.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 7. The input (nA or nB) to output (nB or nA) propagation delays



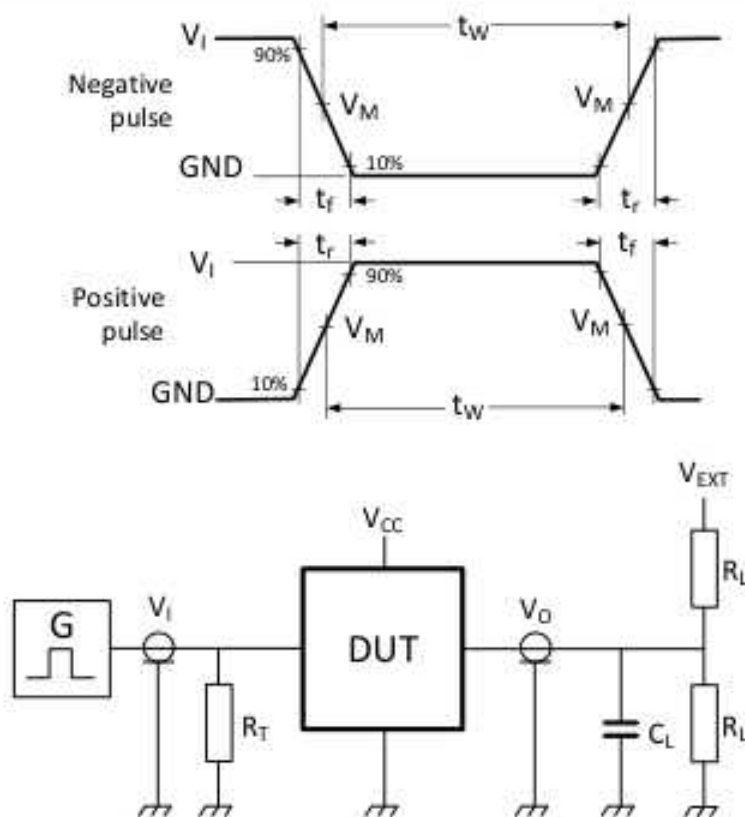
Measurement points are given in Table 9.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

Fig. 8. Enable and disable times

**Table 9. Measurement points**

| Supply voltage | Input       | Output      |                  |                  |
|----------------|-------------|-------------|------------------|------------------|
| $V_{CC}$       | $V_M$       | $V_M$       | $V_X$            | $V_Y$            |
| 2.3 V to 2.7 V | $0.5V_{CC}$ | $0.5V_{CC}$ | $V_{OL} + 0.15V$ | $V_{OH} - 0.15V$ |
| 3.0 V to 3.6 V | $0.5V_{CC}$ | $0.5V_{CC}$ | $V_{OL} + 0.3V$  | $V_{OH} - 0.3V$  |



Test data is given in Table 10.

Definitions for test circuit:

$R_L$  = Load resistance.

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

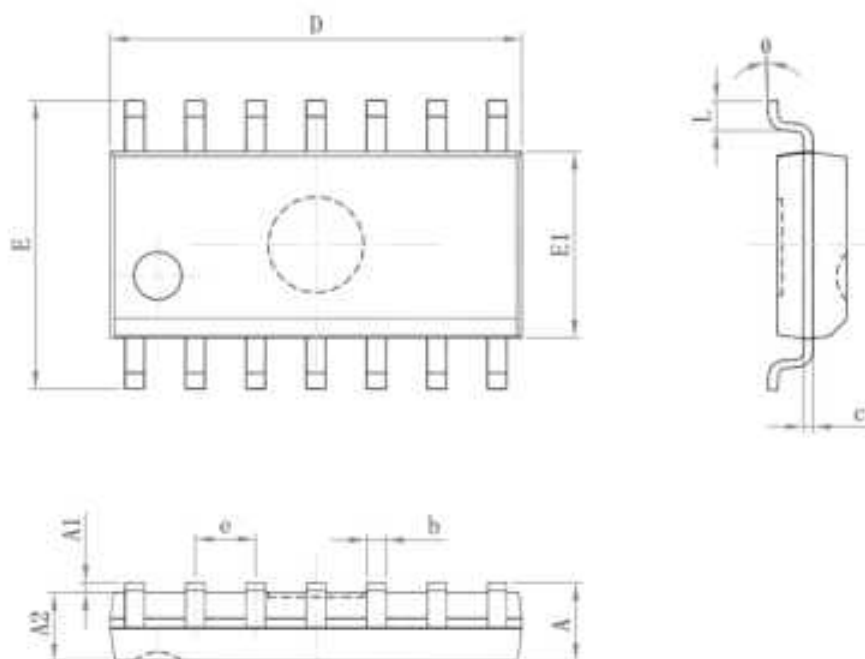
$V_{EXT}$  = External voltage for measuring switching times.

Fig. 9. Test circuit for measuring switching times

Table 10. Test data

| Supply voltage | Input    |               | Load  |              | $V_{EXT}$           |                     |                     |
|----------------|----------|---------------|-------|--------------|---------------------|---------------------|---------------------|
| $V_{CC}$       | $V_I$    | $t_r = t_f$   | $C_L$ | $R_L$        | $t_{PLH1}, t_{PHL}$ | $t_{PZH1}, t_{PHZ}$ | $t_{PZL1}, t_{PLZ}$ |
| 2.3 V to 2.7 V | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ | open                | GND                 | $2 \times V_{CC}$   |
| 3.0 V to 3.6 V | $V_{CC}$ | $\leq 2.0$ ns | 15 pF | 500 $\Omega$ | open                | GND                 | $2 \times V_{CC}$   |

## 11. Package Outline

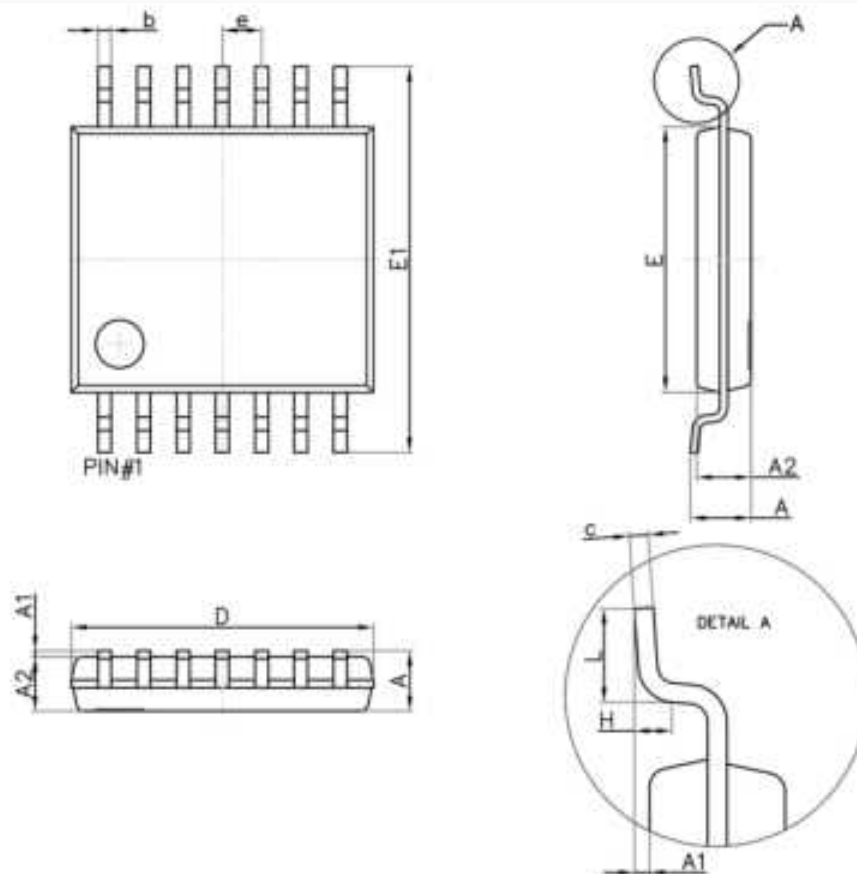
**SOP-14L**


| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min                       | Max   | Min                  | Max   |
| A      | —                         | 1.750 | —                    | 0.069 |
| A1     | 0.100                     | 0.250 | 0.004                | 0.010 |
| A2     | 1.250                     | —     | 0.049                | —     |
| b      | 0.310                     | 0.510 | 0.012                | 0.020 |
| c      | 0.100                     | 0.250 | 0.004                | 0.010 |
| D      | 8.450                     | 8.850 | 0.333                | 0.348 |
| E      | 5.800                     | 6.200 | 0.228                | 0.244 |
| E1     | 3.800                     | 4.000 | 0.150                | 0.157 |
| e      | 1.270(BSC)                |       | 0.050(BSC)           |       |
| L      | 0.400                     | 1.270 | 0.016                | 0.050 |
| θ      | 0°                        | 8°    | 0°                   | 8°    |

# EM74CBTLV3125

4-bit bus switch

## TSSOP-14L



| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| D      | 4.900                     | 5.100 | 0.193                | 0.201 |
| E      | 4.300                     | 4.500 | 0.169                | 0.177 |
| b      | 0.190                     | 0.300 | 0.007                | 0.012 |
| c      | 0.090                     | 0.200 | 0.004                | 0.008 |
| E1     | 6.250                     | 6.550 | 0.246                | 0.258 |
| A      | —                         | 1.200 | —                    | 0.047 |
| A2     | 0.800                     | 1.000 | 0.031                | 0.039 |
| A1     | 0.050                     | 0.150 | 0.002                | 0.006 |
| e      | 0.65 (BSC)                |       | 0.026 (BSC)          |       |
| L      | 0.500                     | 0.700 | 0.020                | 0.028 |
| H      | 0.25(TYP)                 |       | 0.01(TYP)            |       |
| θ      | 1°                        | 7°    | 1°                   | 7°    |

## 12. Tape and Reel Information

### 12.1. Carrier tape dimensions

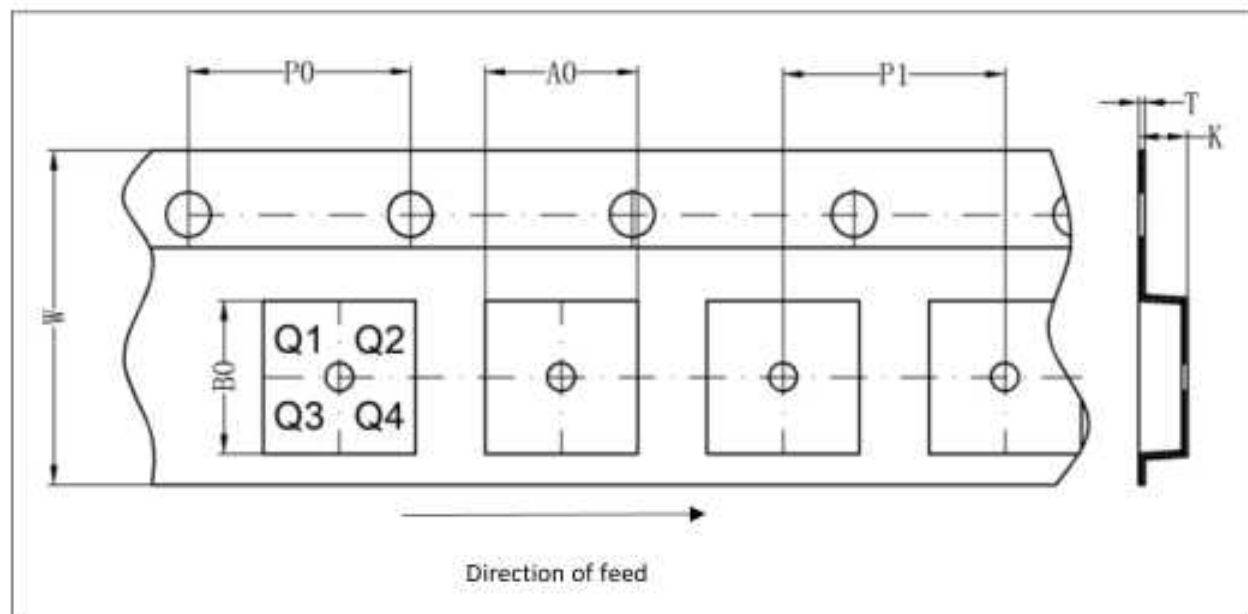
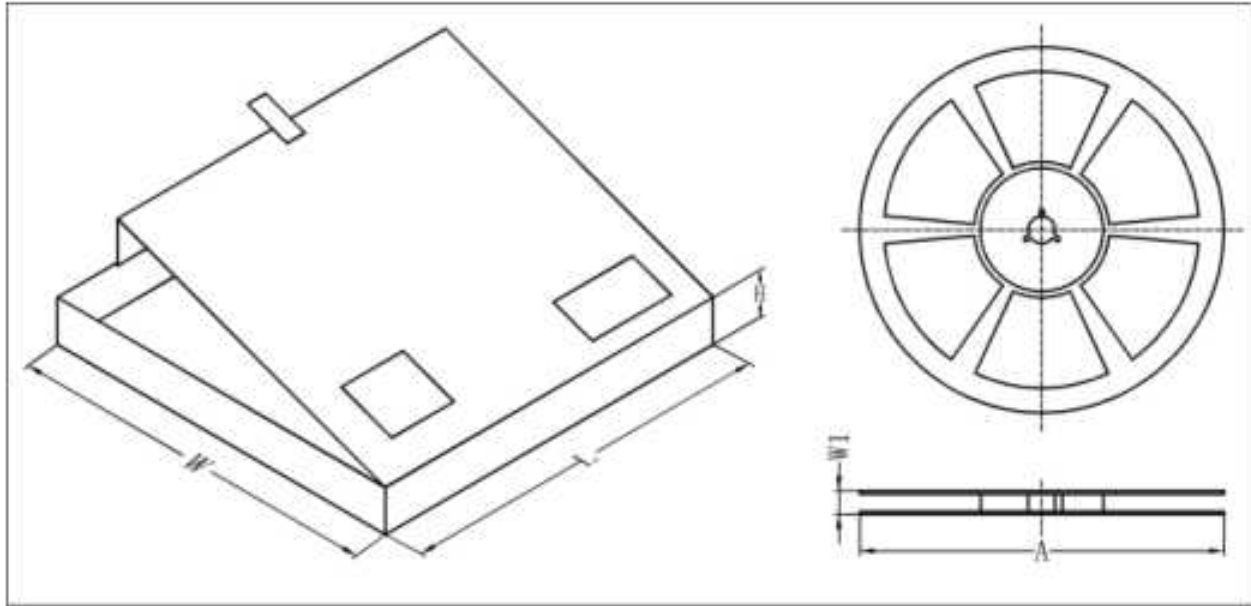


Table 11. Carrier tape dimensions

| Package version | A0(mm) | B0(mm) | K0(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|--------|-------|--------|-------|--------|-------|
| SOP-14L         | 6.35   | 9.1    | 1.9    | 0.22  | 8      | 16    | 4      | Q1    |
| TSSOP-14L       | 6.7    | 5.45   | 1.6    | 0.25  | 8      | 12    | 4      | Q1    |

## 12.2. Reel and box dimensions



**Table 12. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | MPQ (pcs) | Reels per box | Outer box dimensions L×W×H(mm) |
|-----------------|-----------------|-----------------------|--------------------|-----------|---------------|--------------------------------|
| SOP-14L         | D               | 330                   | 22.4               | 3000      | 1             | 358×340×50                     |
| TSSOP-14L       | PW              | 330                   | 18.4               | 3000      | 1             | 358×340×50                     |

## 13. Abbreviations

Table 13. Abbreviations

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | ElectroStatic Discharge                 |
| HBM     | Human Body Model                        |
| CDM     | Charged Device Model                    |
| TTL     | Transistor-Transistor Logic             |

## 14. Revision History

Table 14. Revision history

| Document ID            | Release Date | Data sheet status | Change notice | Supersedes |
|------------------------|--------------|-------------------|---------------|------------|
| EM74CBTLV3125 Rev. 1.0 | Mar 28, 2026 | Product datasheet |               |            |